Adding 2 16 bit numbers stored at 1000 and 1002

1000 and 1001 has one 16 bit number

1002 and 1003 has another 16 bit number

Add and store answer in 1004 and 1005

Carry in 1006

Eg

0x1000 2c (lower order)

0x1001 23 (higher order)

0x1002 f3 (lower order)

0x1003 34 (higher order)

We need to add

232c +

34f3

Program

```
mov r0,#0; to save carry bit
mov dptr,#0x1000; point lower byte
movx a,@dptr; read lower byte
mov b,a; save in a
inc dptr
inc dptr; point lower byte of second number
movx a,@dptr ; read in a
add a,b; add
inc dptr
inc dptr
movx @dptr,a;store in 1004
dec dpl
dec dpl
dec dpl ;point 1001 ie higher byte
movx a,@dptr
```

```
mov b,a
inc dptr
inc dptr
movx a,@dptr; read higher byte of second nmber
addc a,b ; add with carry
jnc next
inc r0
next:
inc dptr
inc dptr
movx @dptr,a; store result
inc dptr
mov a,r0
movx @dptr,a ; store carry
end
```

Check if even nmber

If r0 is even make r1 0 else 1

```
mov r0,#byte
mov a,r0
mov b,#2
mov r1,#1; assume r0 is odd
div ab
mov a,b
cjne a,#0,next
mov r1,#0; r1=0 if even
next:end
```

Using RRC

```
mov r0,#byte
mov a,r0
```

mov b,#2

mov r1,#1; assume r0 is odd

rrc a

jc next

mov r1,#0; r1=0 if even

next:end

Determine factorial

mov r0,#5; to find 5 factorial

mov a,#1

begin:

mov b,r0

mul ab

djnz r0, begin

store a ;will have the factorial

check if given number is prime

mov r0,#byte; to find if byte is prime

mov r1,r0

dec r1

dec r1

mov r2,#0

begin

mov a,r0

mov b,r1

div ab

mov a,b

cjne a,#0,next

inc r2

jmp last

next: djnz r1, begin

last: end ; if r2 is 0 its prime else its not prime